

HOUSTON BUSH, TX, USA

WMO: 722430

Lat: 29.980N

Lon: 95.360W

Elev: 95

StdP: 14.65

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12960

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.4	34.3	16.3	12.6	41.2	21.0	15.8	45.5	23.6	56.1	20.6	53.8	5.8	350	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.2	97.6	76.6	95.7	76.8	93.9	76.8	80.2	88.8	79.5	88.2	79.0	87.6	7.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.3	147.2	82.9	77.5	143.3	82.5	76.9	140.7	82.3	43.7	88.6	43.0	88.3	42.4	87.6	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.3	18.3	16.6	26.3	101.6	3.7	3.2	23.7	103.9	21.5	105.7	19.4	107.5	16.8	109.8		
(5)			23.7	81.9	3.4	1.1	21.3	82.7	19.3	83.3	17.4	83.9	15.0	84.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	70.2	53.6	57.5	63.3	69.7	77.2	82.5	84.4	84.7	80.1	71.6	62.1	55.1
	DBStd	13.19	9.66	9.73	8.40	6.67	5.01	3.47	2.83	3.27	4.61	7.17	8.69	9.54
	HDD50	182	73	37	9	0	0	0	0	0	0	0	11	52
	HDD65	1297	371	245	133	32	1	0	0	0	0	26	154	335
	CDD50	7560	184	245	422	592	842	975	1066	1076	904	669	374	211
	CDD65	3200	17	34	82	174	378	525	601	611	454	230	66	28
	CDH74	32564	51	140	400	1189	3484	5889	7205	7270	4467	1945	418	106
	CDH80	14152	0	14	49	252	1257	2726	3540	3685	1949	617	58	5
Wind	WSAvg	7.3	7.9	8.3	8.6	9.0	8.4	6.8	5.8	5.7	6.0	6.7	7.1	7.4
Precipitation	PrecAvg	49.9	3.5	2.9	3.5	3.6	5.2	5.5	3.7	4.4	4.5	5.2	3.9	3.7
	PrecMax	79.8	9.8	6.1	8.5	14.4	14.4	19.2	12.9	39.1	12.3	16.1	11.7	9.3
	PrecMin	22.9	0.4	0.4	0.1	0.1	0.0	0.1	0.3	0.1	0.8	0.0	0.4	0.6
	PrecStd	11.8	2.2	1.8	2.1	3.0	3.7	4.6	2.6	5.6	3.0	4.1	2.6	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.3	82.1	84.3	88.4	94.2	98.5	99.3	100.8	98.2	91.4	84.8	80.5
		MCWB	66.3	66.8	68.2	71.5	74.3	75.8	76.1	76.5	75.0	74.7	72.5	70.1
	2%	DB	75.2	78.1	81.2	85.2	91.1	95.2	96.8	97.9	94.3	89.0	81.6	77.1
		MCWB	65.1	66.9	68.1	70.5	74.5	76.2	76.8	77.1	75.4	74.4	70.4	68.6
	5%	DB	72.6	75.1	78.6	83.0	88.9	93.2	95.0	95.9	91.9	86.3	78.8	74.1
		MCWB	65.2	66.0	67.3	70.0	74.7	76.5	77.1	77.3	75.5	73.0	69.2	68.1
	10%	DB	69.8	72.1	75.7	80.6	86.8	91.3	93.1	93.7	89.7	83.5	76.0	71.3
		MCWB	63.9	65.5	66.2	69.0	74.0	76.5	77.2	77.4	75.2	71.5	68.6	66.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.5	71.3	72.7	76.0	79.4	81.2	80.8	80.8	80.3	79.2	75.8	72.8
		MCDB	73.3	75.9	78.9	82.6	87.9	88.7	89.7	90.8	87.0	84.3	80.3	76.1
	2%	WB	68.7	69.9	71.2	74.5	78.0	79.9	80.0	80.1	79.2	77.8	73.8	71.1
		MCDB	72.5	74.5	76.6	80.2	85.8	87.7	88.9	89.9	86.6	83.6	78.4	74.9
	5%	WB	67.3	68.6	70.1	73.2	76.9	78.9	79.3	79.4	78.3	76.5	72.0	69.6
		MCDB	71.4	73.1	75.1	78.9	84.3	86.9	88.1	89.0	85.7	82.3	76.7	73.7
	10%	WB	65.2	67.0	68.9	72.1	75.9	78.1	78.6	78.7	77.6	75.1	70.0	67.3
		MCDB	69.3	71.2	73.8	77.6	83.1	86.3	87.3	88.1	85.0	80.7	74.7	70.7
Mean Daily Temperature Range	5% DB	MDBR	19.2	18.8	19.1	18.8	17.8	17.3	17.7	18.2	18.4	19.9	19.8	18.8
		MCDBR	20.8	19.7	19.0	18.8	18.6	19.4	19.8	20.7	20.8	20.3	20.7	19.9
		MCWBR	13.3	11.0	9.3	8.1	6.3	5.1	4.9	5.0	6.3	8.0	11.1	12.9
	5% WB	MCDBR	17.5	16.3	16.2	15.0	15.9	16.2	17.6	18.2	16.4	16.3	17.7	17.7
		MCWBR	13.1	11.6	10.4	8.1	6.6	5.6	5.4	5.3	5.9	8.3	11.9	13.4
Clear-Sky Solar Irradiance	taub	0.339	0.360	0.374	0.436	0.482	0.471	0.498	0.484	0.454	0.380	0.372	0.347	
	taud	2.502	2.454	2.406	2.233	2.143	2.223	2.172	2.213	2.258	2.454	2.425	2.487	
	Ebn at Noon	284	287	289	273	260	261	254	257	261	276	269	275	
	Edh at Noon	29	33	36	45	49	45	48	45	42	32	31	28	
All-Sky Solar Radiation	RadAvg	910	1021	1400	1682	1863	1946	1921	1837	1584	1363	1026	802	
	RadStd	87	182	122	115	137	158	169	145	159	170	105	97	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	+164	+137
(47) Regional (36 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	+200	+175

Nomenclature: See separate page